Unit 4 Comprehensive Review - Mixed - Answer Key

1. $a_1 = 96, \quad a_n = -\frac{1}{2}a_{n-1}$
2. $r = \frac{1}{2}$; $a_n = 32\left(\frac{1}{2}\right)^{n-1}$
3. $x = 4$
4. $r = \frac{1}{4}$; next terms: $4,\ 1,\ \frac{1}{4}$
5. $y = f(x) = (-5)\left(\frac{1}{4}\right)^x$
6. $r = 1.2$; value = 1080
7. y-intercept = 1500; $r = 0.7$; 30% decay
8. $a_n = 3(4)^{n-1}$
9. $a_1 = 4, \quad a_n = 5a_{n-1}$
10. $y = f(x) = (64)(1.5)^x$; at $x = 3$, $y = 216$
11. $y = f(x) = (9)(2)^x$
12. $a_1 = 3200, \quad a_n = 0.75a_{n-1}$
13. $x = \frac{7}{3}$
14. Yes; $r = -3$.
15. y-intercept = 300; $r = 1.12$; 12% growth
16. $a_n = 256\left(\frac{1}{4}\right)^{n-1}$
17. $-2,\ 8,\ -32,\ 128,\ -512$
18. $x = 7$
19. $x = -3$
20. $r = \frac{1}{4}$; $a_1 = 6400, \quad a_n = \frac{1}{4}a_{n-1}$
21. $a_1 = 3, \quad a_n = -2a_{n-1}$
22. $y = f(x) = (45)(3)^{x-2}$; $y = f(x) = (5)(3)^x$
23. $2,\ 8,\ 32,\ 128,\ 512,\ 2048$
24. $y = f(x) = (2500)(0.8)^x$; at $x = 3$, $y = 1280$
25. $r = \frac{2}{3}$; value = 240
26. $r = 5$; next terms: $1250,\ 6250,\ 31250$
27. $125,\ 25,\ 5,\ 1,\ \frac{1}{5}$
28. $x = -3$
29. $x = 2$
30. $a_5 = 486$
31. $r = 2$; $a_n = 7(2)^{n-1}$
32. $a_1 = 6, \quad a_n = 5a_{n-1}$
33. $x = 3$
34. value = 972
35. $1024,\ 256,\ 64,\ 16,\ 4$
36. $a_1 = 7, \quad a_n = 4a_{n-1}$
37. $y = f(x) = (8)(2)^{x-3}$; $y = f(x) = (1)(2)^x$; $y = f(x) = (4)(2)^{x-2}$
38. $x = 5$
39. $r = 3$; $a_n = 9(3)^{n-1}$; $a_6 = 2187$
40. $r = \frac{1}{3}$; $a_n = 729\left(\frac{1}{3}\right)^{n-1}$; $a_6 = 3$
41. $r = 0.5$; value = 150
42. $x = 3$
43. $a_n = 5(-2)^{n-1}$
44. $a_6 = 5$

45. $x = 3$
46. $a_1 = 40, \quad a_n = 1.5a_{n-1}$
47. $r = \frac{1}{3}$; terms: $243,\ 81,\ 27,\ 9,\ 3$
48. $x = 3$
49. $a_1 = 162, \quad a_n = \frac{1}{3}a_{n-1}$
50. No; the consecutive ratios are not constant.
51. $y = f(x) = (5)\left(\frac{1}{2}\right)^{x-2}$; $y = f(x) = (20)\left(\frac{1}{2}\right)^{x}$; $y = f(x) = (10)\left(\frac{1}{2}\right)^{x-1}$
52. $6,\ -18,\ 54,\ -162,\ 486$
53. $x = 2$
54. $a_n = 7(3)^{n-1}$
55. $r = 3$; $a_1 = 8, \quad a_n = 3a_{n-1}$
56. $a_n = 625\left(\frac{1}{5}\right)^{n-1}$